

LECTURE 5

SIMULTANEOUS EQUATIONS IV: LIMITED INFORMATION ML (LIML)

In this lecture, we consider ML estimation of a single equation that is part of a system of simultaneous equations. Without loss of generality, we can focus on the first equation:

$$\begin{aligned} y_{1i} &= X_{1,i}^\top \delta_1 + u_{1i} \\ &= Y_{1,i}^\top \gamma_1 + Z_{1,i}^\top \beta_1 + u_{1i}, \end{aligned}$$

where $Y_{1,i}$ and $Z_{1,i}$ are the vectors of included endogenous and exogenous regressors respectively, as defined in Lecture 2. For the included endogenous regressors we have the following reduced form equation

$$Y_{1,i} = \Pi_1 Z_{1,i} + \Pi_2 Z_{2,i} + V_{1,i}.$$

Note that we ignore $Y_{1,i}^*$, the vector of endogenous variables excluded from the first equation. The two equations above can be written together as

$$\begin{pmatrix} 1 & -\gamma_1^\top \\ 0 & I_{m_1} \end{pmatrix} \begin{pmatrix} y_{1i} \\ Y_{1,i} \end{pmatrix} = \begin{pmatrix} \beta_1^\top & 0 \\ \Pi_1 & \Pi_2 \end{pmatrix} \begin{pmatrix} Z_{1,i} \\ Z_{2,i} \end{pmatrix} + \begin{pmatrix} u_{1i} \\ V_{1,i} \end{pmatrix},$$

or

$$\tilde{\Gamma}_1 \tilde{Y}_{1,i} = \tilde{B}_1 Z_i + \tilde{U}_i,$$

where

$$\begin{aligned} \tilde{\Gamma}_1 &= \begin{pmatrix} 1 & -\gamma_1^\top \\ 0 & I_{m_1} \end{pmatrix}, \\ \tilde{B}_1 &= \begin{pmatrix} \beta_1^\top & 0 \\ \Pi_1 & \Pi_2 \end{pmatrix}, \\ \tilde{Y}_{1,i} &= \begin{pmatrix} y_{1i} \\ Y_{1,i} \end{pmatrix}, \\ \tilde{U}_i &= \begin{pmatrix} u_{1i} \\ V_{1,i} \end{pmatrix}. \end{aligned}$$

Assuming that

$$\tilde{U}_i | Z_i \sim N(0, \tilde{\Sigma}_1),$$

similarly to the derivation of equation (3) in Lecture 4, we obtain that the concentrated log-likelihood for $\tilde{Y}_{1,i}$ is

$$\begin{aligned} Q_n(\tilde{\Gamma}_1, \tilde{B}_1) &= -\frac{(m_1 + 1)}{2} (\log(2\pi) + 1) + \log |\tilde{\Gamma}_1| - \frac{1}{2} \log \left| n^{-1} \sum_{i=1}^n (\tilde{\Gamma}_1 \tilde{Y}_{1,i} - \tilde{B}_1 Z_i) (\tilde{\Gamma}_1 \tilde{Y}_{1,i} - \tilde{B}_1 Z_i)^\top \right| \\ &= -\frac{(m_1 + 1)}{2} (\log(2\pi) + 1) - \frac{1}{2} \log \left| n^{-1} \sum_{i=1}^n (\tilde{\Gamma}_1 \tilde{Y}_{1,i} - \tilde{B}_1 Z_i) (\tilde{\Gamma}_1 \tilde{Y}_{1,i} - \tilde{B}_1 Z_i)^\top \right|, \end{aligned}$$

where the last equality follows from the fact that, due to the restricted structure of $\tilde{\Gamma}_1$, $|\tilde{\Gamma}_1| = 1$.

Thus, LIML is a special case of FIML with properly defined matrices of parameters. However, because of the restricted structure of $\tilde{\Gamma}_1$, there is a closed-form expression for the LIML estimator. Let $\hat{\delta}_1$ be the LIML estimator of $\delta_1 = (\gamma_1^\top, \beta_1^\top)^\top$, then using the matrix notation of Lecture 2, we can write

$$\hat{\delta}_1 = (X_1^\top (I_n - \lambda M) X_1)^{-1} X_1^\top (I_n - \lambda M) y_1,$$

where

$$\begin{aligned} M &= I_n - P, \\ P &= Z(Z^\top Z)^{-1}Z^\top, \end{aligned}$$

and P is the projection matrix onto the space spanned by the exogenous variables Z_i (included and excluded from the first equation),

$$\lambda = \min_t \frac{t^\top W_1 t}{t^\top W t},$$

where

$$\begin{aligned} W &= \begin{pmatrix} y_1 & Y_1 \end{pmatrix}^\top M \begin{pmatrix} y_1 & Y_1 \end{pmatrix}, \\ W_1 &= \begin{pmatrix} y_1 & Y_1 \end{pmatrix}^\top M_1 \begin{pmatrix} y_1 & Y_1 \end{pmatrix}, \end{aligned}$$

and

$$\begin{aligned} M_1 &= I_n - P_1, \\ P_1 &= Z_1(Z_1^\top Z_1)^{-1}Z_1^\top, \end{aligned}$$

and M_1 is the projection matrix onto the space orthogonal to that spanned by $Z_{1,i}$, the exogenous variables included in the first equation. (As defined above, λ is actually the smallest eigenvalue of $W_1 W^{-1}$.)

Next, we will show the asymptotic equivalence of LIML and 2SLS estimators. First, we will show that $\lambda \geq 1$.

$$\begin{aligned} t^\top W_1 t - t^\top W t &= t^\top \begin{pmatrix} y_1 & Y_1 \end{pmatrix}^\top (M_1 - M) \begin{pmatrix} y_1 & Y_1 \end{pmatrix} t \\ &= t^\top \begin{pmatrix} y_1 & Y_1 \end{pmatrix}^\top (P - P_1) \begin{pmatrix} y_1 & Y_1 \end{pmatrix} t. \end{aligned}$$

Since Z_1 is a part of Z , $PZ_1 = Z_1$, and, therefore, $PP_1 = P_1$. Hence,

$$\begin{aligned} (P - P_1)(P - P_1) &= P - P_1 P - P P_1 + P_1 \\ &= P - P_1, \end{aligned}$$

idempotent and, therefore, positive semidefinite. Thus, $t^\top W_1 t - t^\top W t \geq 0$ for any t and $\lambda \geq 1$.

Next, define

$$u_1 = \begin{pmatrix} u_{1i} \\ \vdots \\ u_{1n} \end{pmatrix}.$$

We have

$$\begin{aligned} &\min_t \frac{t^\top W_1 t}{t^\top W t} \\ &= \frac{\begin{pmatrix} 1 & -\gamma_1^\top \end{pmatrix} W_1 \begin{pmatrix} 1 & -\gamma_1^\top \end{pmatrix}^\top}{\begin{pmatrix} 1 & -\gamma_1^\top \end{pmatrix} W \begin{pmatrix} 1 & -\gamma_1^\top \end{pmatrix}^\top} \\ &\leq \frac{\begin{pmatrix} 1 & -\gamma_1^\top \end{pmatrix} \begin{pmatrix} y_1 & Y_1 \end{pmatrix}^\top M_1 \begin{pmatrix} y_1 & Y_1 \end{pmatrix} \begin{pmatrix} 1 & -\gamma_1^\top \end{pmatrix}^\top}{\begin{pmatrix} 1 & -\gamma_1^\top \end{pmatrix} \begin{pmatrix} y_1 & Y_1 \end{pmatrix}^\top M \begin{pmatrix} y_1 & Y_1 \end{pmatrix} \begin{pmatrix} 1 & -\gamma_1^\top \end{pmatrix}^\top} \\ &= \frac{(Z_1 \beta_1 + u_1)^\top M_1 (Z_1 \beta_1 + u_1)}{(Z_1 \beta_1 + u_1)^\top M (Z_1 \beta_1 + u_1)} \\ &= \frac{u_1^\top M_1 u_1}{u_1^\top M u_1}. \end{aligned}$$

Thus,

$$0 \leq \lambda - 1 \leq \frac{u_1^\top (M_1 - M) u_1}{u_1^\top M u_1} = \frac{u_1^\top (P - P_1) u_1}{u_1^\top M u_1}.$$

Lastly,

$$\begin{aligned} n^{1/2} \frac{u_1^\top (P - P_1) u_1}{u_1^\top M u_1} &= \frac{\frac{u_1^\top Z}{n^{1/2}} \left(\frac{Z^\top Z}{n} \right)^{-1} \frac{Z^\top u_1}{n} - \frac{u_1^\top Z_1}{n^{1/2}} \left(\frac{Z_1^\top Z_1}{n} \right)^{-1} \frac{Z_1^\top u_1}{n}}{\frac{u_1^\top u_1}{n} - \frac{u_1^\top Z}{n} \left(\frac{Z^\top Z}{n} \right)^{-1} \frac{Z^\top u_1}{n}} \\ &\rightarrow_p 0, \end{aligned}$$

and, therefore,

$$n^{1/2} (\lambda - 1) \rightarrow_p 0.$$

Next,

$$\begin{aligned} I_n - \lambda M &= I_n - \lambda M + M - M \\ &= I_n - M - (\lambda - 1) M \\ &= P - (\lambda - 1) M. \end{aligned}$$

Hence, the difference between the LIML and 2SLS estimators is given by

$$\begin{aligned} n^{1/2} (\hat{\delta}_1 - \hat{\delta}_1^{2SLS}) &= \left(\frac{X_1^\top (I_n - \lambda M) X_1}{n} \right)^{-1} \frac{X_1^\top (I_n - \lambda M) u_1}{n^{1/2}} - \left(\frac{X_1^\top P X_1}{n} \right)^{-1} \frac{X_1^\top P u_1}{n^{1/2}} \\ &= \left(\frac{X_1^\top P X_1 - (\lambda - 1) X_1^\top M X_1}{n} \right)^{-1} \frac{X_1^\top P u_1 - (\lambda - 1) X_1^\top M u_1}{n^{1/2}} - \left(\frac{X_1^\top P X_1}{n} \right)^{-1} \frac{X_1^\top P u_1}{n^{1/2}} \\ &= \left(\left(\frac{X_1^\top P X_1 - (\lambda - 1) X_1^\top M X_1}{n} \right)^{-1} - \left(\frac{X_1^\top P X_1}{n} \right)^{-1} \right) \frac{X_1^\top P u_1}{n^{1/2}} \\ &\quad - \left(\frac{X_1^\top P X_1 - (\lambda - 1) X_1^\top M X_1}{n} \right)^{-1} \frac{(\lambda - 1) X_1^\top M u_1}{n^{1/2}} \\ &\rightarrow_p 0. \end{aligned}$$